Fleet Route Optimization Report

Generated (UTC): 2026-08-28T16:50:14.564210+00:00 | Use Case: GENERIC
Algorithm: QPSO-VRP | Data Source: FALLBACK TRAFFIC

1. Executive Summary

Total Fleet Distance:	21580.24 km
Total Travel Time:	431.6 hrs
Estimated Fuel Cost:	Rs.172641.92 (assumed Rs.8.0/km)
Traffic Delay Estimate:	N/A (fallback mode)
On-Time Delivery Rate:	0.0%
Vehicles / Customer Stops:	1 vehicles / 10 stops

2. Vehicle Route Manifests

Vehicle #1 - Distance: 21580.24 km | Duration: 431.6 hrs

Seq	Stop Name	Arrival (h)	Leg Dist (km)	Traffic	Status
1	Empire State Building, NY	8.0h	0.0	UNKNOWN	-
2	Delhi (North Hub)	302.32h	14715.89	UNKNOWN	LATE
3	Jaipur (Pink City)	308.19h	293.66	UNKNOWN	LATE
4	Varanasi (Central UP)	326.69h	925.09	UNKNOWN	LATE
5	Visakhapatnam (East Port)	347.83h	1056.62	UNKNOWN	LATE
6	Chennai (South Port)	362.82h	749.61	UNKNOWN	LATE
7	Bangalore (Tech Hub)	370.08h	363.18	UNKNOWN	LATE
8	Hyderabad (Central Hub)	382.52h	622.08	UNKNOWN	LATE
9	Pune (Auto Hub)	395.18h	632.86	UNKNOWN	LATE
10	Mumbai (West Port)	398.18h	150.17	UNKNOWN	LATE
11	Kolkata (East Hub)	439.6h	2071.08	UNKNOWN	LATE

3. Traffic & Congestion Analysis

Segment Breakdown: High Congestion: 0 | Medium: 0 | Low: 0
Note: Fallback free-flow traffic model used. Figures assume ideal road conditions.

4. Optimization Performance

Total Iterations: 100 | Initial Cost: 11653890.0 | Final Optimal Cost: 10449145.79
Convergence Rate: None | Quantum Tunneling Events: 182 | Runtime: 4.0836s

5. Recommendations & Operational Notes

- Total distance, time, and cost figures above summarize this specific optimization run and are based on the data source noted in metadata.

- Compare the optimization_performance section across repeated runs to gauge consistency of the solver on your data.

- Data source for this report: fallback traffic data, as noted in metadata.data_source. Fallback-mode figures assume free-flow conditions and will understate real-world delay.